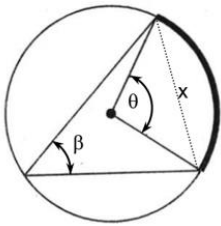


Continuation... Terms related to a circle

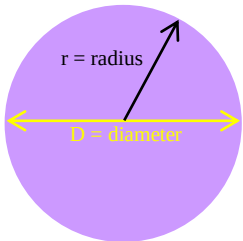
- Central angle** – an angle whose vertex is at the center of a circle and whose sides are the radii. The central angle is an angle subtended by an arc.
- Angle subtended by a chord** – an angle whose vertex is along the periphery or circumference and its sides are chords.



The angle θ is the central angle while β is the angle subtended by chord "x".

Formulas:

- Area and Circumference of a circle



$$\text{Area} = \pi r^2$$

$$\text{Area} = \frac{\pi D^2}{4}$$

$$\text{Circumference} = 2\pi r = \pi D$$

- Sector of a circle



$$A = \frac{1}{2}rc$$

$C = \text{arc length}$

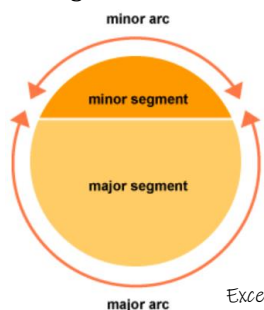
$$A = \frac{1}{2}r^2\theta$$

θ in radians

$$A = \frac{\pi r^2 \theta}{360}$$

θ in degrees

- Segment of a circle



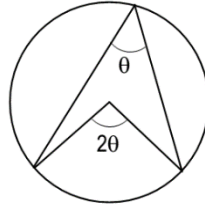
For minor segment:
Area = area of circle – area of sector

For major segment:
Area = area of circle – area of minor segment

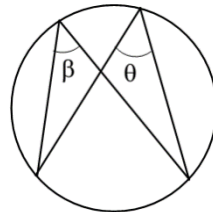
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Theorem involving a circle

- If a central angle and a peripheral angle are subtended by the same arc, then the central angle is twice as large as the peripheral angle.

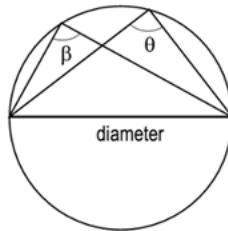


- Inscribed angles subtend the same arc are equal.



$$\theta = \beta$$

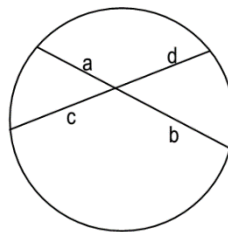
- Inscribed angles subtended by the diameter of a circle are right angles.



$$\theta = \beta = 90^\circ$$

The triangle formed in the figure is a right triangle

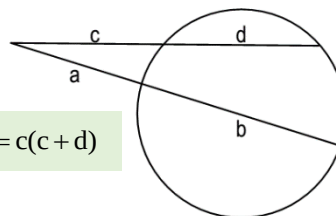
- Chord Theorem



$$ab = cd$$

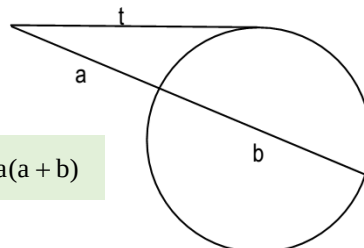
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- Secant Theorem



$$a(a+b) = c(c+d)$$

- Secant-Tangent Theorem

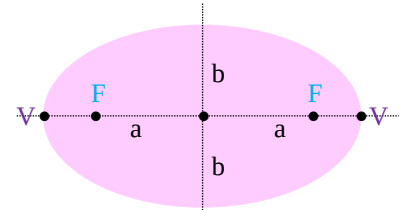


$$t^2 = a(a+b)$$

Ellipse

An ellipse is a locus of a point which moves so that the sum of its distances to the fixed points (foci, denoted as F in the figure) is constant and is equal to the length of the major axis (2a).

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Ellipse has two axis of symmetry, a major axis (2a), between the vertices (V) of the ellipse, and a minor axis (2b), which intersect at the center of the ellipse.

Formulas:

$$\text{Area} = \pi ab$$

$$\text{Perimeter} = 2\pi \sqrt{\frac{a^2 + b^2}{2}}$$



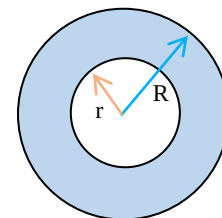
The formula for the circumference or perimeter of an ellipse shown above is just an approximate formula.

Annulus

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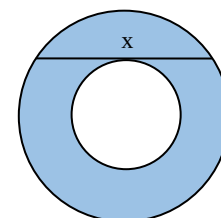
An annulus is a ring-shaped object, a region bounded by two concentric circles.

- Given radii (R and r) of the two circles



$$\text{Area} = \pi(R^2 - r^2)$$

- Given length of tangent chord



$$A = \frac{\pi x^2}{4}$$



Did You Know?

To date, more than half the total number of Electronic Engineers in the Philippines reviewed at Excel Review Center!